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School Segregation During Adolescence is Associated with Higher 30-Year Cardiovascular Risk of Black but not White Young Adults

Kristi L. Allgood[1,2] · Nancy L. Fleischer[2] · Shervin Assari[3] · Jeffrey Morenoff[4,5] · Belinda L. Needham[2]

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Abstract
Although cardiovascular disease (CVD) mortality rates are declining for American adults, a disparity remains between non-Hispanic Black and non-Hispanic White adults. Previous research has shown that residential segregation, a form of structural racism, experienced in childhood is associated with later-life racial and ethnic health disparities, including disparities in CVD and its risk factors. However, little is known about the health consequences of exposure to segregated schools, especially among those living in neighborhoods with high concentrations of minoritized people. This study used data from the In-School, Wave I, and Wave IV surveys of the National Longitudinal Study of Adolescent to Adult Health to examine a novel school measure of school racial segregation (Index of the Concentration of Extremes, ICE) as a predictor of Framingham 30-year CVD risk scores. We used General Estimating Equation models to evaluate the association between ICE, measured at Wave I, and two different 30-year CVD risk scores, measured at Wave IV, and examined whether the relationship varied by race. We observed that higher levels of school segregation were associated with a higher 30-year CVD risk among non-Hispanic Black participants while higher segregation was associated with a lower 30-year CVD risk for non-Hispanic White participants. This research suggests that exposure to segregation in the school setting may contribute to observed disparities in CVD among US adults.

Keywords Racialized school segregation · Cardiovascular risk · Health disparities · Index of concentration of extremes

Introduction

Although the cardiovascular disease (CVD) mortality rates for both Black and White adults have declined over time, the mortality gap between Black and White Americans has increased, with a 40% higher risk of CVD mortality among Black compared to White Americans in 2020 [1]. Indeed, these disparities remain even after accounting for individual factors such as age, gender, and socioeconomic status, as well as traditional risk factors such as diabetes, smoking, and hypertension [2, 3]. Additionally, research suggests that stressful exposures in childhood contribute to higher CVD risk later in life [4, 5]. Many researchers have argued that a contributing factor affecting the health and well-being of racially minoritized groups is structural racism and racialized residential segregation in particular [6–11].

Residential Segregation

Structural racism is defined as structures or barriers that are created by interconnecting and reinforcing policies, laws, and/or customs that produce racially disparate outcomes where racially minoritized group members are negatively affected and White persons are privileged [6, 8, 12–17]. Some policies were created with discriminatory intent and produced structures and institutions that explicitly excluded or isolated racial and ethnic minoritized groups. Examples

✉ Kristi L. Allgood
kallgood@tamu.edu

1 Department of Epidemiology and Biostatistics, Texas A&M University, 212 Adriance Lab Road, Office 231, College Station, TX 77843, USA

2 School of Public Health, Department of Epidemiology, Center for Social Epidemiology and Population Health, University of Michigan, Ann Arbor, MI, USA

3 Department of Family Medicine, Charles R. Drew University of Medicine and Science, Los Angeles, CA, USA

4 Department of Sociology, University of Michigan, Ann Arbor, MI, USA

5 Institute for Social Research, Population Health Studies, University of Michigan, Ann Arbor, MI, USA

of explicitly racist policies are Jim Crow (1896–1964), which legalized all types of racialized segregation in 21 US states plus Washington DC, and the Homeowners Loan Act of 1933, which legalized discriminatory mortgage lending nationwide [18–22]. Indeed, racialized residential segregation was federally legal from at least 1865 through 1977 when the US Congress passed the Community Reinvestment Act, which legally mandated that local banks provide services that meet the needs of all community members, regardless of race [23, 24]. By the time of this act and beyond, however, racialized residential segregation was persistent in most American communities [24].

Segregation in Schools

Racialized residential segregation leads to segregation in other social domains such as education [25, 26]. Educational racialized segregation is commensurate with residential segregation because where people live dictates where they or their children can attend public school [24]. Shortly after the end of the American Civil War, state-specific racialized segregation laws were enacted, and eventually, the legality of the state laws was argued in front of the US Supreme Court which codified racial segregation in public spaces in southern states, including schools, created through *Plessy v Ferguson* or "Separate but Equal" in 1896 [27]. While school and residential segregation are still ongoing, the legally racist school segregation policy ended with the 1954 *Brown v Board of Education of Topeka, Kansas* Supreme Court decision, which began the desegregation of southern schools with court-ordered oversight [26, 28]. The *Brown v Board* ruling made no mention of schools in the northern states; consequently, schools and communities in the northern urban US remained more segregated than urban areas in the southern states through at least 2020 [24, 26, 28, 29]. In response to the *Brown* ruling, private schools catering to White students, referred to as "segregationist academies," began to open. These schools were clustered in the rural south until at least 1975 and contributed at least somewhat to contemporary school segregation [30]. Additionally, the 1964 Civil Rights Act (CRA) did not further improve school segregation, as a portion of this law did not include the *means* to reduce school segregation [31, 32]. For example, the CRA stated that schools could not utilize federal funds for bussing or any transportation to bring racially and ethnically minoritized students to schools educating predominantly White students [32, 33]. To combat the slow response to the desegregation court orders, the federal government provided financial incentives to desegregate more rapidly [34]. The additional policies after the CRA did push schools to desegregate, and by the 1970s, US public schools were far less segregated than prior years [35]. Indeed, there were other Supreme Court cases that furthered the desegregation process by restricting "Freedom of Choice" student assignment plans, mandating strict racial quotas, bussing (for schools that could fund it locally), and continued court-ordered oversight of desegregation [26, 36, 37]. Scholars have found that these were effective methods to desegregate schools, while also improving the education of both racially minoritized and White students [26, 38]. However, in the early 1990s, the Supreme Court ruled to release school districts from desegregation oversight, and the schools quickly began resegregating because they returned to assigning students to schools based on the racially segregated neighborhoods in which they lived [26, 39].

After desegregation oversight ended in the 1990s, educational quality declined resulting in growing Black vs. White gaps in achievement and lower high school completion rates by race [40–43], and some suggest that educational quality is worse in segregated schools in highly minoritized neighborhoods than it was prior to the *Brown* decision [24, 26]. Indeed, educational resources prior to desegregation were flowing to public schools, especially in the South, the achievement gaps between Black and White students improved as did other metrics of educational quality (e.g., student-to-teacher ratios, high school completion) [40, 42]. Residential segregation creates concentrated poverty [44]; as a result, schools are segregated by both race and poverty status, which can prevent educators from providing a quality education because teachers then must focus on broader social issues in the classroom such as hunger, fear, stress, crime, behavioral issues, and neglect [24]. This differential quality of basic education results in fewer racially minoritized students being able to attend college thereby creating another barrier to economic success [45]. Many argue that the differential quality of education is driving this disparity, at least partially [24, 46, 47].

Thus far, there has been a small handful of studies examining racialized segregation in schools in relation to health outcomes such as behavioral problems in youth [48], adolescent mental health [49, 50], adolescent body mass index [51], midlife cognitive health [35, 52, 53], depression [54], substance use [54], poor self-rated health [54], and some CVD risk factors [55]. These studies suggest that segregation in schools is deleterious to the physical and mental health of the students while in school [48–51], but also later in life [35, 52–56]. Using a quasi-experimental approach and adjusting for residential segregation, Kim et. al. found that attending segregated schools in 1991 was associated with reporting poor health, binge drinking, and cardiovascular disease by 2017 among respondents who attended schools that were once under court-ordered desegregation during their adolescent years [55]. Additionally, Walsemann and colleagues observed that in highly segregated schools of former Jim Crow states, using primary school attendance records, differences in school term policy and self-reported

attendance explained about half of the Black to White racial disparity in cognitive function in older ages [56]. Collectively, these studies suggest that school segregation may be, at least in part, responsible for some of the observed disparities in health outcomes such as CVD, independent of residential segregation.

In the present study, we examined racial differences in the association between school racial segregation during adolescence and individual-level 30-year CVD risk among US non-Hispanic Black and non-Hispanic White adults. Using data from the National Longitudinal Study of Adolescent to Adult Health (Add Health), we hypothesized that the association between school segregation at Wave I, when respondents were between 7 and 12th grade, and 30-year CVD risk at Wave IV, when respondents were between 24 and 32 years old, would differ between non-Hispanic Black and non-Hispanic White adults. Specifically, we hypothesized that among non-Hispanic Black adults, attending racially segregated schools would be associated with a higher 30-year CVD risk score, whereas among non-Hispanic Whites, attending racially segregated schools would be associated with a lower 30-year CVD risk score. This hypothesis stems from common definitions of structural racism where policies, norms, and customs create barriers to progress for Black persons and other racially and ethnically minoritized persons while simultaneously privileging White persons [6, 8, 12–17, 57–59].

Methods

The design of this study was a population-based, multi-level, cohort study using nationally representative data. The main exposure was school segregation resulting from past racist federal policies [24]. The operationalization of structural racism was through the school-level measure of school segregation using the Index of Concentration of Extremes based on the race of students in schools. The outcome of this study was the Framingham 30-year CVD risk score [60].

The primary data source was the National Longitudinal Study of Adolescent to Adult Health (Add Health). The Add Health methodology has been extensively published elsewhere [61–63]. Briefly, Add Health is a nationally representative school-based prospective cohort study of adolescents that examines behavioral, emotional, social, educational, and contextual factors as respondents transition to adulthood. The baseline sample was gathered beginning in 1995 when participants were in 7th–12th grade using a complex clustered sampling design. The initial sampling frame was a list of schools from the Quality of Education Data, Inc. from which 80 high schools and their accompanying feeder middle schools were randomly selected. All students in selected schools were provided a screening survey (i.e., In-School Survey) and served as a sampling frame for the cohort participants (i.e., Wave I and IV surveys). Students in selected schools were stratified based on grade and sex, and 17 students within each stratum were selected with additional samples taken for Black children with at least one parent with a college degree, students with disabilities, and siblings. The Wave I survey included 20,745 adolescents who were representative of American adolescents with respect to region, urbanicity, school features, and ethnicity. Subsequent Waves II–V followed up with participants who completed Wave I. Field interviews were gathered in the participants' homes. Topics included demographics, socioeconomic status, physical health, and risk behaviors. In addition to data collected through interviews, height and weight were measured at Wave I, and objective laboratory measurements (i.e., blood pressure, measured height and weight) were also taken starting in Wave IV using systematic data collection protocols.

Data for this study were from the In-School survey, the Wave I survey, the Wave I contextual data file, and the Wave IV survey. The primary exposure of racialized school segregation was based on data measured prior to Wave I using the data from the In-School survey. The outcome was measured at Wave IV, when the cohort was between 24 and 32 years old. By Wave IV, personal habits and chronic diseases are beginning to affect Add Health participants. Eighty percent of respondents who completed Wave I completed the Wave IV survey.

Measurement of School Segregation

The main exposure in this study was racialized school segregation. The school segregation measure is called the Index of Concentration of Extremes (ICE), which measures the extent to which a school is over-populated by extremes in racial concentration.

The formula for the Index of Concentration of Extremes (ICE) [58, 64] is as follows:

$$ICE = \frac{(Number\ of\ NH - White\ students\ in\ an\ area) - (Number\ of\ NH - Black\ students\ in\ an\ area)}{Total\ population\ in\ the\ school\ with\ information\ on\ race}$$

All students in randomly selected schools were provided with an In-School survey during a school day (n=90,118). This survey included questions on demographics and eligibility for the primary and ancillary Add Health studies, including student's race and ethnicity. Race and ethnicity were collected as a series of binary (yes/no) questions. In order to obtain a count

of students by race for each school, an overall race variable was created by combining all race and ethnicity questions provided into one variable where those reporting Hispanic were coded in one category (regardless of reported race), then those reporting their race as Black, White, or another race were coded in accordance with the Office of Management and Budget's (OMB) guidance on race and ethnicity [65]. This created a race and ethnicity variable of Hispanic, non-Hispanic Black, another non-Hispanic race, and non-Hispanic White groups for the In-School Survey participants. Then, each level of this 4-level variable was recoded into a binary indicator that was coded as 1 if the respondent reported that racial and ethnic group and 0 if they did not. This coding scheme created four new binary variables each representing a different racial and ethnic group in the study and were subsequently aggregated up to the school level to calculate the number of non-Hispanic Black and non-Hispanic White students within each school. The ICE denominator included all students in the school with non-missing information on race ($n = 2760$ had missing data on race in the school survey). The final ICE value ranged from − 1 to 1, where negative values indicated a higher concentration of Black students, and positive values indicated a higher concentration of White students in schools. Numbers closer to − 1 or + 1 indicated a higher level of school segregation based on race.

Outcomes

Given the young adult age range for the Add Health participants in Wave IV, a low incidence of cardiovascular disease (CVD) in younger adults, and a lack of medical record review to confirm self-reported CVD events, we examined the Framingham CVD risk score measured using Wave IV data as the primary outcome. The risk score, which was developed using data from the Framingham Heart Study, is a useful clinical tool that predicts the risk of CVD events occurring within the next 30 years. To calculate the risk score, we used the procedure developed by Pencina and colleagues [60], which predicts the risk of a set of cardiovascular events occurring using a Cox proportional hazards model that accounts for competing causes of death. The Cox model included the following covariates: age, sex, use of antihypertensive medications, smoking status, diabetes status, systolic blood pressure, and body mass index. The procedure produces two risk scores, each accounting for a different set of CVD outcomes. The risk score has been previously applied to Add Health data [66, 67]. The first 30-year CVD risk score predicts the risk of the following outcomes occurring within 30 years: coronary death, myocardial infarction, and fatal and non-fatal stroke (hard CVD risk). The second risk score (full CVD risk) predicts those outcomes in the hard CVD risk score plus the following: coronary insufficiency, angina pectoris, transient ischemic attack, intermittent claudication, and congestive heart failure [60].

Covariates

Covariates were selected a priori and are associated with both school segregation and 30-year CVD risk. The following individual covariates were included: age, sex, race, parental education, and parental income. *Age* was based on the age of the respondent at Wave IV (Supplemental Figure S1). *Sex* and *race* were self-reported at Wave I. Each respondent was allowed to report multiple races and/or ethnicities. The respondent *race* was coded following the OMB [65] which defined racial categories for the US Census (described above). The research question to be answered involves the effect of federal housing policies that, at the time they were enacted, targeted Black Americans; as such, we limit the sample to those who report their race as non-Hispanic Black and non-Hispanic White. *Parental education* and *parental income* were based on responses from the parents at Wave I. Parental education was defined as the highest level of education for the respondent's most highly educated parent; categories include less than high school, high school graduate, college graduate, and unknown. Parental income was defined in the following categories < \$20,000, \$20,000–\$39,999, \$40,000–\$74,999, ≥ \$75,000, and unknown and is based on parental income at Wave I. Unknown categories were included for parental education and income to maintain sample size. This study examined race as an effect modifier to determine whether the relationship between school segregation and CVD risk differed by race.

Additionally, several tract-level covariates were included in the models measured at Wave I (1994–1995) using the Add Health contextual data file (Supplemental Figure S1). Tract-level covariates include percent Hispanic residents, percent Black residents, percent of adults with less than a high school diploma, urbanicity, and the percent of residents below the federal poverty level (FPL). Tract-level covariates were measured at Wave I (1994–1995) using the 1990 US Census values for the tract of respondent residence. Income inequality was represented as the ICE by tract level median household income (ICE_{Income}) as defined by Krieger et al. and Feldman et al. [58, 64]. ICE_{income} was calculated using income levels to determine financial privilege with similar methods as described for the exposure variable for this study. In this formula, the high-income group was defined as those with incomes at or above the 80th percentile of all incomes in 1990, which was \$55,205, and the low-income group was defined as those with incomes at or below the 20th percentile of all incomes in 1990, which was \$12,500.[1] These exact income values were not available in the Add

[1] https://www2.census.gov/programs-surveys/cps/tables/time-series/historical-income-households/h01ar.xlsx

Health dataset; thus, those in the tract earning $50,000 or more were defined as the high-income group, and those in the tract earning $15,000 or less were defined as the low-income group. The denominator includes all persons in the tract for whom income was available.

Analytic Sample

After merging the In-School, Wave 1, the Wave I contextual data, and Wave IV surveys and deleting those records that did not have a Wave I weight value, per the analytic guidelines, there were 12,887 individuals included in the dataset. Of this, there were 9205 individuals who reported their race as non-Hispanic Black or non-Hispanic White. These racial groups were selected because they are the two predominant racial groups that have been affected by residential segregation in the United States, which has contributed to the disadvantage of non-Hispanic Blacks relative to non-Hispanic Whites in the US. Finally, participants without complete data on all variables were excluded from the analysis. The final sample for this study was $n = 6839$. The sample size difference is made up of those missing tract level information because their addresses could not be geocoded (0.9%), those missing race from the In-School survey (22%), and those missing data on covariates or the outcome (4%) (see Supplemental Figure S2 and Table S3). A domain analysis was conducted which means that that although respondents with missing data were excluded from the analyses, they were coded out of the analytic sample to preserve the original complex data structure. The domain analysis is recommended by the Add Health analytic guidelines.

Statistical Analysis

Both outcome variables were substantially right-skewed and were subsequently log-transformed. Table 1 displays all variables included in the models. To determine the statistical difference between the non-Hispanic Black and Non-Hispanic White adults, the *t*-test was used to examine group means for continuous variables and the modified Rao-Scott chi-square test to examine differences in proportions.

We used a Generalized Estimating Equations (GEE) analytic approach to examine the population average effect of racial school segregation on 30-year CVD risk [68]. GEE was performed using PROC GENMOD (SAS version 9.4) with an identity link (and normal distribution) and a repeated statement that accounts for clustering at the school level and strata (region) variables. Grand sampling survey weights were employed to account for the complex survey design, non-response, and post-stratification.

For the analytic strategy, Model 1 included school segregation only. Model 2 added the individual and area-level covariates. Then, to determine if the association between school segregation and the population average 30-year CVD risk score varied by race, an interaction between the school segregation exposure and race was added (Model 3). Additionally, we ran race-stratified models to estimate the differences observed by racialized group membership. As a sensitivity analysis, to determine if the observed disparity was affected by residential segregation, we additionally adjusted for a commonly used measure of tract-level residential segregation called the Index of Dissimilarity, which measures the extent to which Black or White residents in a tract would have to move to achieve an even racial distribution in the tract [69]. All models were run separately for each risk score. The SAS GENMOD procedure does not have an option to produce a domain analysis; thus, for those participants who reported their race as something other than non-Hispanic Black or non-Hispanic White, the grand sampling weight was recoded to 0.0001 per the analytic instructions [62]. This allows for the complex structure of the data to remain intact, while also excluding the groups who are outside of the study population and is consistent with the Add Health analytic guidelines [62].

All analyses were conducted using survey packages available in SAS 9.4. Interpretation of the log-transformed variables can be examined by taking the anti-log of the coefficients produced in the models, which result in a geometric mean (similar to a median) value for that variable. The interpretation for the anti-log of the interaction term is the increased or decreased risk using the following equation $[(\exp^{coefficient})-1]*100$, where an anti-log > 1 suggests an increased 30-year CVD risk and an anti-log < 1 suggests a decreased risk. Statistical significance was determined at the $p = 0.05$ level.

Results

Eighty-seven percent of respondents reported that their parents had at least a high school diploma, and 11% reported parental incomes of greater than $75,000 (Table 1). The school segregation value overall was 0.35, which indicates that there is a moderate level of school racial segregation in the communities represented (i.e., positive values of the school segregation indicator suggest a higher concentration of White students relative to Black students whereas negative values indicate a higher concentration of Black students relative to White students). The average income inequality was − 0.064, which indicates that there was more income disadvantage relative to income advantage in the tracts represented in the Add Health study. Racial differences in socioeconomic status were observed. White respondents had higher parental incomes compared to Black respondents ($p < 0.001$). Black

Table 1 Descriptive statistics overall and by race, Add Health Wave I ($n = 6839$)

Measure	Mean or %	Overall	Non-Hispanic Blacks		Non-Hispanic Whites		*p*-value[a]
		$N = 6839$	$N = 1990$		$N = 4849$		
		95% CI	Mean or %	95% CI	Mean or %	95% CI	
Race (%)							
Non-Hispanic Black	18.8	(13.4, 24.2)	—	—	—	—	—
Non-Hispanic White	81.2	(75.8, 86.7)	—	—	—	—	—
Age at Wave IV (yrs)	28.1	(27.8, 28.5)	28.4	(28.0, 28.7)	28.1	(27.8, 28.3)	0.2058
Sex (%)							
Female	52.3	(50.6, 54.0)	43.3	(39.7, 46.9)	48.7	(49.4,53.2)	0.0795
Male	47.7	(46.0, 49.5)	56.7	(53.1, 60.3)	51.3	(46.8, 50.6)	*Ref*
Parental education (%)							
< HS	8.2	(6.6, 9.7)	11.7	(7.9, 15.5)	7.3	(5.9, 8.8)	0.4052
HS graduate	52.4	(49.8, 55.0)	53.3	(48.6, 58.0)	52.1	(49.1, 55.1)	0.5161
College grad	35.1	(31.8, 38.4)	27.0	20.8, 33.2)	37.0	(33.3, 40.6)	*Ref*
Unknown	4.4	(3.6, 5.1)	8.0	(6.2, 9.8)	3.6	(2.9, 4.3)	**0.0063**
Parental income (%)							
< $20,000	14.7	(12.3, 17.1)	27.7	(21.8, 33.7)	11.7	(9.5, 14.0)	**< 0.0001**
$20,000–$39,999	23.3	(21.5, 25.1)	24.6	(21.2, 28.1)	23.0	(21.0, 25.0)	0.3985
$40,000–$74,999	31.4	(28.8, 34.0)	16.7	(13.1, 20.3)	34.8	(32.1, 37.5)	**0.0003**
$75,000 or more	11.4	(9.4, 13.3)	5.4	(3.8, 7.0)	12.7	(10.5, 15.0)	*Ref*
Unknown	19.2	(17.4, 21.0)	25.6	(20.5, 30.6)	17.8	(16.0, 19.6)	**0.0001**
Outcomes							
30-year CVD risk (full)[b]	0.123	(0.119, 0.127)	0.135	(0.127, 0.142)	0.120	(0.116, 0.124)	**0.0026**
30-year CVD risk (hard)[b]	0.068	(0.065, 0.070)	0.080	(0.071, 0.083)	0.066	(0.063, 0.068)	**0.0018**
Exposure							
ICE (race in school)	0.346	(0.291, 0.401)	-0.076	(-0.152, 0.000)	0.444	(0.403, 0.484)	**< 0.0001**
Tract-level covariates							
Income inequality (ICE)—Tract	-0.064	(-0.096, -0.033)	-0.240	(-0.288, -0.192)	-0.0001	(-0.056, 0.009)	**< 0.0001**
% in tract with < HS	26.8	(25.3, 28.4)	35.1	(32.8, 37.3)	24.9	(23.3, 26.6)	**< 0.0001**
Urban tract, % Yes	39.9	(31.3, 48.6)	49.6	(34.4, 64.8)	37.7	(28.5, 47.0)	0.4456
% Latinx/Hispanic in tract	3.1	(2.5, 3.6)	3.2	(2.3, 4.2)	3.0	(2.4, 3.6)	0.7134
% Black in tract	14.8	(11.2, 18.3)	54.9	(49.0, 60.8)	5.4	(4.2, 6.8)	**< 0.0001**
% below federal poverty level in tract	14.2	(12.7,15.7)	25.4	(22.7, 28.1)	11.6	(10.3, 12.9)	**< 0.0001**

Ref reference value, *HS* high school, *ICE* Index of Concentration of Extremes, *FPL* federal poverty level

[a] *p*-values are generated through *t*-test for continuous variables and the chi-square test for categorical/binary variables. Bolded *p*-values indicate statistical significance at the 0.05 level

[b] 30-year cardiovascular disease risk score predicts the risk of the following outcomes occurring within 30 years: coronary death, myocardial infarction, fatal and non-fatal stroke (hard) + coronary insufficiency, angina pectoris, transient ischemic attack, intermittent claudication, and congestive heart failure (full). The risk score is calculated using a cox-proportional hazards model including the following variables in the prediction model: sex, blood pressure, hypertension medication use, diabetes status, smoking status, and body mass index (BMI); see Pencina et. al., [60] for methods

respondents had higher unadjusted cardiovascular disease risk compared to White respondents ($p = 0.003$ for the full CVD outcome, and $p < 0.001$ for the hard CVD outcome). Finally, Black respondents attended schools with a higher concentration of Black students whereas White respondents attended schools with a higher concentration of White students ($p < 0.001$).

Full 30-Year CVD Risk

As shown in Model 1 of Table 2, school segregation was not associated with a full 30-year CVD risk. After controlling for hypothesized confounding variables, the coefficient for school segregation remained negative and non-significant, and the coefficient for race was not statistically significant (Model 2). Model 3 revealed a

Table 2 Regression[a] for 30-year full cardiovascular risk score (log-transformed) at Add Health Wave IV on the Index of Concentration of Extremes at the School (School Based Segregation) at Wave I ($n = 6839$)

Measure	Log-transformed full 30-year CVD risk score[b]								
	Model 1[c]			**Model 2[c]**			**Model 3[c]**		
	Beta	**95% CI**	***p*-value**	**Beta**	**95% CI**	***p*-value**	**Beta**	**95% CI**	***p*-value**
Intercept	−2.283	(−2.333, −2.232)	**< 0.0001**	−5.492	(−5.727, −5.247)	**< 0.0001**	−5.479	(−5.724, −5.234)	**< 0.0001**
Index of Concentration of Extremes (school)	−0.048	(−0.163, 0.067)	0.4124	−0.017	(−0.095, 0.061)	0.6718	−0.056	(−0.146, 0.034)	0.2224
Race									
Black	—			0.032	(−0.022, 0.086)	0.2444	0.012	(−0.046, 0.069)	0.6886
White (ref)	—			—			—		
Age at Wave 4 (yrs)	—			0.094	(0.086, 0.102)	**< 0.0001**	0.094	(0.086, 0.102)	**< 0.0001**
Sex									
Male	—			0.672	(0.639, 0.706)	**< 0.0001**	0.672	(0.639, 0.706)	**< 0.0001**
Female (ref)	—			—			—		
Parental education									
< HS	—			0.197	(0.119, 0.274)	**< 0.0001**	0.196	(0.118, 0.273)	**< 0.0001**
HS graduate	—			0.095	(0.054, 0.136)	**< 0.0001**	0.094	(0.053, 0.135)	**< 0.0001**
College grad (ref)	—			—			—		
Unknown	—			0.110	(0.024, 0.196)	**0.0118**	0.108	(0.022, 0.193)	**0.0138**
Parental income									
< $20,000	—			0.107	(0.039, 0.174)	**0.0020**	0.109	(0.042, 0.176)	**0.0015**
$20,000–$39,999	—			0.117	(0.059, 0.175)	**< 0.0001**	0.118	(0.061, 0.176)	**< 0.0001**
$40,000–$74,999	—			0.042	(−0.009, 0.092)	0.1047	0.043	(−0.007, 0.093)	0.0887
$75,000 or more (ref)	—			—			—		
Unknown	—			−0.037	(−0.016, 0.089)	0.1738	0.038	(−0.015, 0.090)	0.1580
Income inequality (ICE) in tract	—			−0.036	(−0.151, 0.080)	0.5428	−0.031	(0.146, 0.085)	0.6030
Urban tract (yes)	—			0.045	(0.006, 0.084)	**0.0255**	0.039	(−0.001, 0.078)	0.0550
% Black in tract	—			−0.091	(−0.200, 0.018)	0.1009	−0.066	(−0.180, 0.049)	0.2600
% Latinx/Hispanic in tract	—			−0.255	(−0.467, −0.044)	**0.0179**	−0.298	(−0.518, −0.078)	**0.0080**
% < HS in tract	—			0.231	(0.039, 0.423)	**0.0183**	0.239	(0.052, 0.425)	**0.0121**
% below FPL in tract	—			0.293	(−0.017, 0.603)	0.0639	0.292	(−0.015, 0.600)	0.0626
ICE (School)*race	—			—			0.159	(0.010, 0.308)	**0.0362**
Exchangeable correlation	0.019			0.002			0.002		

[a]General Estimating Equations (GEE) with the identity link

[b]30-year full cardiovascular disease risk score predicts the risk of the following outcomes occurring within 30 years: coronary death, myocardial infarction, and fatal and non-fatal stroke, coronary insufficiency, angina pectoris, transient ischemic attack, intermittent claudication, and congestive heart failure. The risk score is calculated using a cox-proportional hazards model including the following variables in the prediction model: sex, blood pressure, hypertension medication use, diabetes status, smoking status, and body mass index (BMI); see Pencina et. al., [60] for methods

[c]Model 1 includes exposure alone; Model 2 includes Models 1 + race, age, sex, parental education, parental income, income inequality, urbanicity, % Black, % Latinx/Hispanic, % with < HS, % below federal poverty level (FPL); Model 3 includes Models 1–2 and a race*exposure interaction

Bolded *p*-values indicate statistical significance at the 0.05 level; *FPL* federal poverty level

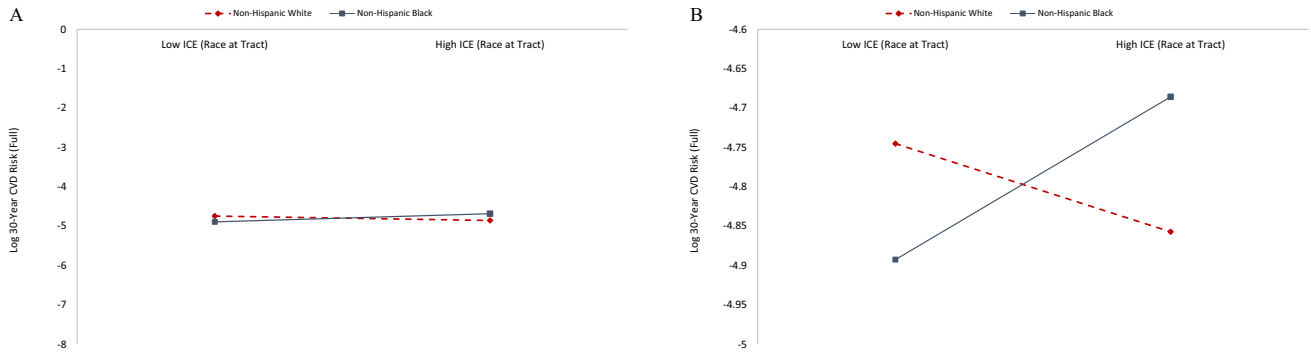

Fig. 1 Estimated Wave IV Log — 30-year full cardiovascular disease (CVD) for low and high residential segregation (Index of Concentration of Extremes at school) at Wave I by race, Add Health. **A** Original graphic with axis set with 0 as the high value. **B** Zoomed in to see the relationship between the slopes. [a]General Estimating Equations (GEE) with the identity link, *p*-value for interaction = **0.0362** (see Table 2). Interpretation: for each unit increase in Index of Concentration of Extremes (school), there is an increase in CVD risk of 11% among Blacks and a decrease of 6% for Whites. [b]30-year full cardiovascular disease risk score predicts the risk of the following outcomes occurring within 30 years: coronary death, myocardial infarction, fatal and non-fatal stroke, coronary insufficiency, angina pectoris, transient ischemic attack, intermittent claudication, and congestive heart failure. The risk score is calculated using a cox-proportional hazards model including the following variables in the prediction model: sex, blood pressure, hypertension medication use, diabetes status, smoking status, and body mass index (BMI); see Pencina et. al., [60] for methods. [c]Model includes exposure, race, age, sex, parental education, parental income, income inequality, urbanicity, % Black, % Latinx/Hispanic, % with < HS, % below federal poverty level (FPL)

statistically significant race*school segregation interaction (coefficient for interaction = 0.159, $p = 0.036$). A higher concentration of White students in the school was associated with a higher CVD risk score for Black respondents but a lower CVD risk score for White respondents (see Fig. 1). Indeed, Black respondents had an approximate 11% *higher* risk of fatal and non-fatal CVD events occurring within the next 30 years for each unit increase in school segregation, while White respondents had an approximate 6% *lower* risk of fatal and non-fatal CVD events occurring within the next 30 years for each unit increase in school segregation. In stratified models (Table 4), similar findings were observed and suggested that the predicted mean 30-year CVD risk score for Black participants was 10.8%, while it was 9.9% for White participants. This resulted in a small but statistically significant difference (Model 2: mean difference 0.092, 95% CI 0.059, 0.125, $p < 0.0001$). In a sensitivity analysis, results were similar after the addition of the Index of Dissimilarity as a measure of residential segregation, except in Model 2 where the non-significant coefficient for school segregation was positive (rather than negative) (Supplemental Table S1 and Supplemental Figure S3).

Hard 30-Year CVD Risk

School segregation was not associated with hard 30-year CVD risk (Models 1 and 2 in Table 3). Model 3 revealed a statistically significant race*school segregation interaction ($p = 0.048$). The interaction can be observed in Fig. 2 where CVD risk was higher for Black respondents but lower for White respondents as the concentration of White students in the school increased. Indeed, Black respondents had an approximate 13% *higher* risk of a fatal or non-fatal CVD event occurring within the next 30 years for each unit increase in school segregation measured by ICE, while White respondents had an approximate 6% *lower* risk of a fatal or non-fatal CVD event occurring within the next 30 years each unit increase in school segregation. In stratified models (Table 4), similar findings were observed and suggested that the predicted mean 30-year CVD risk score for Black participants was 5.5%, while it was 4.9% for White participants. This resulted in a small but statistically significant difference (Model 2: mean difference 0.109, 95% CI 0.069, 0.148, $p < 0.0001$). With the addition of the Index of Dissimilarity, similar results were observed except in Model 2 where the coefficient for school segregation was positive (rather than negative) but remained non-significant (Supplemental Table S2). Also like the main analysis, the interaction between school segregation and race was statistically significant (Supplemental Table S2, Model 3 and Supplemental Figure S4).

Discussion

This study examined the relationship between a measure of school segregation and two Framingham 30-year CVD risk scores in a nationally representative sample of young adults. The primary hypothesis was that higher levels of school segregation would be associated with higher 30-year CVD

Table 3 Regression[a] coefficients for 30-year hard cardiovascular risk score (log-transformed) at Add Health Wave IV on the Index of Concentration of Extremes at the school (school-based segregation) at Wave I ($n = 6839$)

Measure	Log-transformed hard 30-year CVD risk score[b]								
	Model 1[c]			**Model 2[c]**			**Model 3[c]**		
	Beta	**95% CI**	***p*-value**	**Beta**	95% CI	***p*-value**	**Beta**	**95% CI**	***p*-value**
Intercept	−2.977	(−3.037, −2.917)	**< 0.0001**	−6.689	(−6.977, −6.402)	**< 0.0001**	−6.675	(−6.962, −6.387)	**< 0.0001**
Index of Concentration of Extremes (school)	−0.057	(−0.194, 0.081)	0.4211	−0.018	(−0.110, 0.075)	0.7099	−0.063	(−0.169, 0.043)	0.2415
Race									
Black	—			0.048	(−0.018, 0.112)	0.1519	0.024	(−0.044, 0.092)	0.4922
White (ref)	—			—			—		
Age at Wave 4 (yrs)	—			0.107	(0.097, 0.116)	**< 0.0001**	0.107	(0.097, 0.116)	**< 0.0001**
Sex									
Male	—			0.896	(0.857, 0.936)	**< 0.0001**	0.896	(0.857, 0.936)	**< 0.0001**
Female (ref)	—			—			—		
Parental Education									
< HS	—			0.233	(0.139, 0.327)	**< 0.0001**	0.232	(0.138, 0.327)	**< 0.0001**
HS graduate	—			0.111	(0.064, 0.159)	**< 0.0001**	0.110	(0.062, 0.158)	**< 0.0001**
College grad (ref)	—			—			—		
Unknown	—			0.132	(0.030, 0.233)	**0.0110**	0.129	(0.027, 0.230)	**0.0129**
Parental income									
< $20,000	—			0.122	(0.043, 0.200)	**0.0025**	0.124	(0.046, 0.203)	**0.0019**
$20,000– $39,999	—			0.131	(0.063, 0.200)	**0.0002**	0.133	(0.065, 0.201)	**0.0001**
$40,000– $74,999	—			0.044	(-0.015, 0.102)	0.1460	0.046	(−0.013, 0.104)	0.1252
$75,000 or more (ref)	—			—			—		
Unknown	—			0.041	(−0.020, 0.103)	0.1852	0.043	(−0.018, 0.104)	0.1683
Income inequality (ICE) in tract	—			−0.043	(−0.178, 0.093)	0.5380	−0.036	(−0.171, 0.098)	0.5968
Urban tract (yes)	—			0.050	(0.005, 0.096)	**0.0304**	0.044	(−0.003, 0.090)	0.0638
% Black in tract	—			−0.104	(−0.236, 0.029)	0.1241	−0.074	(−0.213, 0.064)	0.2945
% Latinx/Hispanic in tract	—			−0.287	(−0.544, −0.031)	**0.0281**	−0.337	(−0.605, −0.070)	**0.0135**
% < HS in tract	—			0.250	(0.026, 0.475)	**0.0291**	0.259	(0.041, 0.477)	**0.0200**
% below FPL in tract	—			0.338	(−0.020, 0.696)	0.0645	0.337	(−0.019, 0.693)	0.0638
ICE (school)*race	—			—			0.185	(0.002, 0.369)	**0.0476**
Exchangeable correlation	0.017			0.002			0.002		

[a]General Estimating Equations (GEE) with the identity link

[b]30-year hard cardiovascular disease risk score predicts the risk of the following outcomes occurring within 30 years: coronary death, myocardial infarction, fatal & non-fatal stroke. The risk score is calculated using a Cox-Proportional Hazards model including the following variables in the prediction model: sex, blood pressure, hypertension medication use, diabetes status, smoking status, and body mass index (BMI); see Pencina et. al., [60] for methods

[c]Model 1 includes exposure; Model 2 includes Models 1 + race, age, sex, parental education, parental income, income inequality, urbanicity, % Black, % Latinx/Hispanic, % with < HS, % below federal poverty level (FPL); Model 3 includes Models 1–2 and a race*exposure interaction

Bolded *p*-values indicate statistical significance at the 0.05 level; *FPL* federal poverty level

risk scores among Black respondents but lower risk scores for White respondents. Despite a non-statistically significant association in the adjusted model between school segregation and 30-year CVD risk, we observed a statistically significant interaction between school segregation and race, with results in the hypothesized directions.

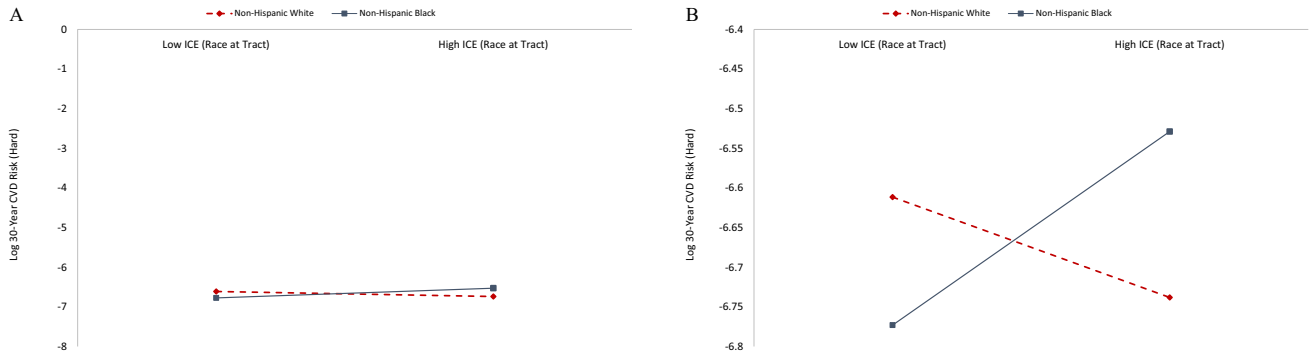

Fig. 2 Estimated Wave IV Log — 30-year hard cardiovascular disease (CVD) for low and high residential segregation (Index of Concentration of Extremes at school) at Wave I by race, Add Health. **A** Original graphic with axis set with 0 as the high value. **B** Zoomed in to see the relationship between the slopes. [a]General Estimating Equations (GEE) with the identity link, *p*-value for interaction = 0.0476 (see Table 3). Interpretation: for each unit increase in Index of Concentration of Extremes (tract), there is an increase in CVD risk of 13% among Blacks and a decrease of 6% for Whites. [b]30-year hard cardiovascular disease risk score predicts the risk of the following outcomes occurring within 30 years: coronary death, myocardial infarction, and fatal and non-fatal stroke. The risk score is calculated using a cox-proportional hazards model including the following variables in the prediction model: sex, blood pressure, hypertension medication use, diabetes status, smoking status, and body mass index (BMI); see Pencina et. al., [60] for methods. [c]Model includes exposure, race, age, sex, parental education, parental income, income inequality, urbanicity, % Black, % Latinx/Hispanic, % with < HS, % below federal poverty level (FPL)

Table 4 Predicted means and mean differences for 30-year cardiovascular risk score (log-transformed) at Add Health Wave IV on the Index of Concentration of Extremes at the school (school-based segregation) at Wave I ($n = 6839$) stratified by race and ethnicity

	Log of 30-year cardiovascular risk score (full outcomes)		
	Predicted means		Difference
	Non-Hispanic Black	Non-Hispanic White	Black–White
	$N = 1990$	$N = 4845$	
Model 1	−2.227 (−2.282, −2.172)	−2.313 (−2.352, −2.275)	0.087 (0.025, 0.148), ***p* = 0.0061**
Anti-log (geometric mean)	0.108	0.099	
Model 2	−2.222 (−2.250, −2.194)	−2.314 (−2.333, −2.295)	0.092 (0.059, 0.125), ***p* < 0.0001**
Anti-log (geometric mean)	0.108	0.099	
Sensitivity (with IoD), Model 2	−2.222 (−2.250, −2.194)	−2.314 (−2.333, −2.296)	0.092 (0.060, 0.125), ***p* < 0.0001**
Anti-log (geometric mean)	0.108	0.099	
	Log of 30-year cardiovascular risk score (hard outcomes)		
	Predicted means		Difference
	Non-Hispanic Black	Non-Hispanic White	Black–White
	$N = 1990$	$N = 4845$	
Model 1	−2.910 (−2.978, −2.841)	−3.014 (−3.059, −2.968)	0.104 (0.028, 0.180), ***p* = 0.0072**
Anti-log (geometric mean)	0.054	0.049	
Model 2	−2.907 (−2.941, −2.872)	−3.015 (−3.034, −2.994)	0.109 (0.069, 0.148), ***p* < 0.0001**
Anti-log (geometric mean)	0.055	0.049	
Sensitivity (with IoD), Model 2	−2.907 (−2.941, −2.872)	−3.016 (−3.038, −2.994)	0.109 (0.070, 0.149), ***p* < 0.0001**
Anti-log (geometric mean)	0.055	0.049	

Bolded *p*-values indicate statistical significance at the 0.05

In this study, we observed an association between a higher 30-year CVD risk for Black participants, but a lower 30-year CVD risk for White students. Notably, by the start of the Add Health Survey in the mid-1990s, segregation was well entrenched in American society including in schools due to the US Supreme Court ending desegregation oversight [26]. This is also despite residential segregation being outlawed for nearly 20 years through federal policies that banned discrimination in the sale and rental of housing, mortgage lending, and redlining in predominantly Black neighborhoods [70]. Our results are consistent with the broader literature on residential segregation and health outcomes, in general, where a higher level of residential segregation is more strongly associated with adverse birth outcomes [71–87],

cancer outcomes [88–90], HIV mortality [91], both fatal and non-fatal cardiovascular outcomes [92, 93], hypertension [58], survival [94], fatal police shootings [95], and community violence among Black persons compared to White persons [64, 96, 97]. Additionally, some research suggests that higher levels of residential segregation is also harmful to White adults with respect to CVD risk factors and outcomes, which may depend on poverty status [92, 98, 99].

A unique component of this study was the use of a novel school segregation measure as an indicator of structural racism. In this context, school segregation was defined as Black students being more likely to attend school with Black students while White students were more likely to attend school with White students. We observed that the association between school segregation and 30-year CVD risk score was dependent on race and more harmful to Black participants than to White participants for both the full and hard CVD risk scores, even after accounting for residential segregation. This finding is consistent with other research on school segregation and non-cardiovascular disease outcomes such as mental health, substance abuse, and self-rated health [48–50, 54].

School and residential segregation and housing policies are two mechanisms of how structural racism is reinforced and shaped over time. Given that polices can affect one's access to resources that are health-promoting and that policies are written in a way to limit certain populations' access to resources, it is plausible that a potential mechanism by which school segregation can affect health is through a material pathway, such as educational attainment, income, or housing [6, 16]. Another plausible mechanism operates through a psychosocial pathway through perceived injustice, social status, and stress [6]. For example, because of racism that is experienced every day in some fashion, even at young ages including in utero, Black people accumulate the effects of stress, and without adequate resources, at schools and at home, over time, the effect of accumulated stressors eventually takes its physical toll [100–103]. The stressor of anticipating racial discrimination (i.e., vigilance), particularly among Black students attending schools with high concentrations of White students, is associated with higher levels of hypertension, which may explain why Black students who attended schools with a higher concentration of White students had worse CVD risk in early adulthood in our study [104]. Indeed, both anticipating racial discrimination and high-effort coping (i.e., John Henryism) when faced with discrimination may be additional stressors that Black students face, particularly when they are attending schools with a high concentration of White students [105, 106].

Limitations and Strengths

A potential limitation of this study is that there could be some selection bias due to non-participation. This non-participation may be related to the outcome as well as the exposure, where those who are more disadvantaged, due possibly to strained resources from neighborhood disinvestment, may be less likely to participate. To address this potential limitation, the Add Health data is adjusted for non-response and post-stratification. Indeed, after examining measured confounding variables across various groupings of potentially eligible participants, there were no meaningful differences in any variables (Supplementary Table S3). There were a substantial number of respondents for which race was unavailable in the In-School survey, the sampling frame for the Add Health cohort. These cases were excluded. In the full dataset, there were 9205 respondents who were non-Hispanic Black or non-Hispanic White. The final sample size was 6839 respondents. If these missing cases are related to the exposure or outcome this may present a source of bias [107]. There was also some item non-response for parental socioeconomic status. We attempted to mitigate this potential bias by coding an unknown category for income and education, which allowed us to maintain some of the sample sizes for missing covariates such as parental education and income; however, we could not recover racial and ethnic groups of students in the In-School survey. Additionally, we examined the demographic variables for all records in Add Health, those where respondents reported their race and non-Hispanic Black or non-Hispanic White, and those who were in the analytic sample (Supplemental Table S3). There was consistency between those reporting non-Hispanic Black and non-Hispanic White race and those who were included in the analytic sample across all factors suggesting that the drop in sample size for the analytic sample did not affect our results. Finally, we could not examine potential stress-related biological mechanisms as measures of perceived stress were only available in later waves. This is an important area for future research but beyond the scope of this study.

Despite these limitations, there are several strengths of note. Add Health is a long-term and nationally representative cohort of young adults in the US that includes both contextual factors and individual responses. Temporality is established in Add Health as school segregation is measured at Wave I while the outcome is measured at Wave IV. While it is possible that CVD processes begin in high school, older adolescents have not accumulated the needed risk factors to develop chronic CVD, so it is unlikely that respondents at Wave I had CVD at the time of the first interview. Another major strength is the use of a novel measure of school segregation as a proxy for structural racism that captures some of the most longstanding and disadvantaging policy areas in US history [24]. Index of Concentration of Extremes is an easily calculated measure and has yet to be applied to school settings, making it a novel measure of segregation. Finally, the Add Health data include physical and laboratory measurements, which improve the accuracy of the outcome measurement.

Conclusion

The present study builds on a small body of research on school segregation and later life health outcomes. Using a novel measurement of school segregation we examined the association between school segregation and risk for CVD later in life. We observed that, consistent with other studies, the relationship between school segregation and CVD risk is dependent on race and that school segregation is associated with a higher 30-year CVD risk for Black young adults, who endure disparate impacts of historic policies. This study adds to the literature that examines area-level indicators of structural racism as predictors of individual health outcomes by incorporating a novel measure of structural racism (segregation in schools). Examining segregation in schools is important because higher-quality education provides better economic opportunities as one ages. Segregated schools in neighborhoods with high concentrations of minoritized people have been shown to have lower educational quality for several reasons such as concentrated poverty, inadequate and racially disparate resources [108], fear, anxiety, and hunger [24]. The implications of this study suggest that understanding the role of school segregation, in excess of residential segregation, on CVD risk factors that place racial and ethnic minoritized populations at a disadvantage for the health of the cardiovascular system is critical for eliminating cardiovascular disease disparities.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s40615-024-02135-5.

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Author Contribution Conceptualization: Kristi L. Allgood, Belinda L. Needham; methodology: Kristi L. Allgood, Belinda L. Needham, Nancy L. Fleischer; formal analysis and investigation: Kristi L Allgood; writing — original draft preparation: Kristi L. Allgood; writing — review and editing: Belinda. L. Needham, Nancy L. Fleischer, Shervin Assari, Jeffrey Morenoff; supervision: Belinda L. Needham.

Data Availability Data are publicly available through the Carolina Population Center at the University of North Carolina at Chapel Hill, https://addhealth.cpc.unc.edu/data/. Code for the Framingham Cardiovascular Disease Risk Score is available upon request and was provided by Dr. Pencina at Duke University and Ken Williams at KenAnCo Biostatistics.

Declarations

Ethics Approval This study fell under an umbrella of Institutional Review Board approval at the University of Michigan for the use of secondary data with the Add Health dataset.

Competing Interests The authors declare no competing interests.

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